

Global exponential stability and H_∞ control of limit cycle for switched affine systems under time-dependent switching signal

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ABSTRACT

In this paper, the stability analysis of continuous-time switched affine systems (CTSASs) is addressed via dwell time switching. The main purpose of this note is to design a time-dependent switching signal assuring the global exponential stability of a desired limit cycle (which is selected from a set of attainable limit cycles). By constructing a discretized Lyapunov function and using the linear matrix inequalities technique, a set of conditions in the framework of dwell time are designed. For CTSASs, the performance indexes of the system in this neighborhood cannot be analyzed because most of the proposed controllers drive the system trajectory into a small enough neighborhood. Differently, the technique proposed in this note leads the CTSAS's trajectory to the limit cycle so that the resulting conditions can consider the weighted H_∞ performance level of the system. More specifically, the design conditions can be applied to open-loop subsystems that are unstable. At last, a numerical example and a practical example are given to illustrate our method.

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1. Introduction

The switched system, which is constituted by some subsystems and a switching law that decides which subsystem is activated, the theory of its research involves filtering [1,2], control [3–9] and other fields [10–14]. In a switched linear system, the state trajectories of each subsystem can converge to the equilibrium point over time. If there exists an affine term in the switched system, equilibrium points of switched subsystems can't be moved to the origin, so that the switched system has several equilibria in the state space, which is a prominent feature of the switched affine system. Since there are many nonlinear complex systems can be modeled by switching affine systems (SASs), SASs have been widely used in industrial engineering fields such as power converters [15–18], network system control [19], spacecraft [20], traffic planning [21] and biology [22]. The controlling idea for a switched affine system is to formulate a switching rule that can guide the system trajectory from any starting coordinate to the expected equilibrium points (which is selected from a set of attainable equilibrium points).

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For a continuous-time switched affine system (CTSAS), some literature has designed state feedback or dynamic output feedback controllers utilizing conditions based on a quadratic Lyapunov function [15,23–25] or max-type Lyapunov function [26]. However, stability in the above work requires high-frequency switching, which is not possible in the real physical world. When the switching frequency is constrained, the asymptotic stability of the system cannot be guaranteed, but the practical stability. The idea to guarantee the practical stability of the CTSAS is to guide the state trajectory of the system to the convergence region including the equilibrium points. Some important results on the practical stability of CTSASs are given in [27–32]. The recent lecture [27] introduces a positive dwell time to ensure the practical stabilization for CTSASs to avoid sliding modes and generate arbitrarily fast switching, and the dwell time of it can be adjusted by the design parameters. In literature [28,29], practical stability analysis for switched affine systems based on the use of sampled-data controllers are studied. The work [30] studies practical stability of CTSASs under minimum dwell time criteria while the work [33] under average dwell time criteria. **Since the controllers proposed in all those references drive the system trajectory into a small enough region, the system behavior within this region cannot be analyzed and controlled**, which results in the H_∞ performance index can't be taken into account. **Some recent works [34,35] have considered the local stabilization of a limit cycle for the CTSAS through a systematic approach based on a Poincaré map method.** The very recently papers [36,37] have investigated the global asymptotic stability of a limit cycle for CTSASs and discrete-time switched affine systems (DTSASs), respectively. It is worth mentioning that, up to date, there are few studies on the stability of limit cycles for CTSASs, no results coping with global exponential stability (GES) of a limit cycle for CTSASs. Meanwhile, all the literatures mentioned in this paragraph adopts state-dependent switching signal. In general, the time-dependent switching signal (TDSS) is more convenient to be applied in practical projects because the system does not need to obtain the state information all the time.

This paper proposes a TDSS to ensure the GES of the desired limit cycle for the CTSAS. At the same time, the H_∞ control is considered to ensure the proper performance of the transient response, which a situation that cannot be taken into account in actual stability. The TDSS obtained in this paper is more convenient to be applied in practical projects because it does not need to obtain the full state information of the switched affine system all the time like the state-dependent switching function. Also, the design conditions can be applied to open-loop subsystems that are unstable. Finally, a numerical example and a practical example show the validity of the obtained results of this technical note.

The structure of this paper is shown as follows. System and limit cycle descriptions are given in Section 2. Section 3 presents the main theorems include stability analysis and TDSS design for CTSASs. Two numerical simulation examples used to verify the main results of this paper are provided in Section 4. Finally, Section 5 gives the conclusion.

Notations: There are standard notations used in this paper. The superscript "T" denotes matrix transposition. $\mathbb{N}$ and $\mathbb{N}^+$ represent the set of non-negative and positive integers, respectively. $\mathbb{R}$ stands for the set of real numbers. $\mathbb{R}^n$ represents the n -dimension Euclidean vector space.

2. System and limit cycle descriptions

The CTSAS considered in this paper are constructed as follows:

$$\begin{aligned}\dot{x}(t) &= A_{\sigma(t)}x(t) + B_{\sigma(t)}w(t) + a_{\sigma(t)}, x(0) = x_0, \\ y(t) &= C_{\sigma(t)}x(t) + D_{\sigma(t)}w(t),\end{aligned}\quad (1)$$

where $x(t) \in \mathbb{R}^n$ denotes the system state vector, $w(t) \in \mathbb{R}^m$ denotes the disturbance input and $y(t) \in \mathbb{R}^p$ represents the measured output. $\sigma: \mathbb{N}^+ \rightarrow \mathcal{I} = \{1, 2, \dots, N\}$ is a TDSS that selects at each instant of time one of the N available subsystems. A_i, B_i, C_i, D_i, a_i are known real matrices and for simplicity of the i th ($\forall i \in \mathcal{I}$) subsystem. The sequence of switching instants $\{t_{s_1}, t_{s_2}, t_{s_3}, \dots\}$ is increasing and does not exist any accumulation.

Remark 1. From the expression of system (1), we can see that when $a_i = 0$ for all $i \in \mathcal{I}$, system (1) will becomes a continuous-time linear switched system.

Suppose the system (1) with a periodic equilibrium solution $x_e(t)$, where the period is defined as $T > 0$. The equilibrium solution $x_e(t)$ with a periodic switching sequence $\sigma(t) = c[k(t)] = (c(0), c(1), \dots, c(\kappa - 1))$, $\forall t \in \mathbb{R}$ satisfies

$$\dot{x}_e(t) = A_{c[k(t)]}x_e(t) + a_{c[k(t)]}, \quad (2)$$

with the correspondent limit cycle

$$\mathcal{X}_e(c) = \{x_e(t) : (2), t \in \mathbb{R}\}. \quad (3)$$

Let $\tau_{k(t)}$ is the dwell time of mode $c[k(t)]$, so the period $T = \sum_{i=0}^{\kappa-1} \tau_i$. Notice that points of $x_e(t)$ can be obtained from (2) for $t \in [0, T]$ respecting the boundary condition $x_e(0) = x_e(\kappa - 1)$. The switching signal is determined by the N^κ to be selected periodic switching sets $\sigma(t) = c[k(t)]$ which is based on criteria related to the expected steady-state behavior of the system state. Figure 1 visually shows the relationship between the above symbols. Notice that the range of values for T and $\tau_{k(t)}$ will be given in the next section.

In this paragraph, the objective is to confirm the trajectories $x_e(t)$ associated with the switching signal $\sigma(t) = c[k(t)] \in \mathcal{I}$. For the given period of the limit cycle T , let $\mathcal{C}(\kappa) = \mathcal{I}^\kappa$ be the set given by the Cartesian product of $\mathcal{I}$ with itself κ times. This set includes N^κ elements $c[k(t)] \in \mathcal{C}(\kappa)$, which are finite sequences $c = (c(0), c(1), \dots, c(\kappa - 1))$. Associated to each

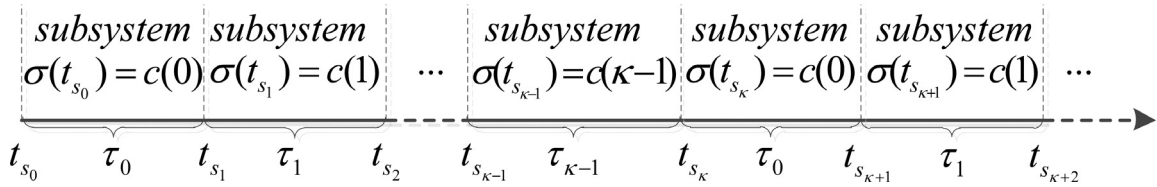
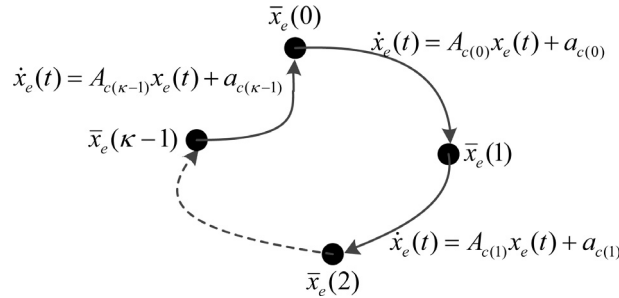


Fig. 1. An illustration of several key symbols.

Fig. 2. Limit cycle with n key nodes of a switched affine system.

$c \in \mathcal{C}(\kappa)$ there exists $x_e(c)$ given in (3), whose points of $x_e(t)$ can be obtained from (2) with $x_e(0) = x_e(T)$ as a solution to the following equation

$$\bar{A}(c)\bar{x}_e = -\bar{a}(c), \quad (4)$$

where $\bar{x}_e = [\bar{x}_e^T(0) \quad \bar{x}_e^T(1) \quad \cdots \quad \bar{x}_e^T(\kappa-1)]^T$ is the key node in $x_e(t)$,

$$\bar{A}(c) = \begin{bmatrix} e^{A_{c(0)}\tau_0} & -I & 0 & \cdots & 0 \\ 0 & e^{A_{c(1)}\tau_1} & -I & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -I & 0 & 0 & \cdots & e^{A_{c(\kappa-1)}\tau_{\kappa-1}} \end{bmatrix}, \quad \bar{a}(c) = \begin{bmatrix} \int_0^{\tau_0} e^{A_{c(0)}t} dt a_{c(0)} \\ \int_0^{\tau_1} e^{A_{c(1)}t} dt a_{c(1)} \\ \vdots \\ \int_0^{\tau_{\kappa-1}} e^{A_{c(\kappa-1)}t} dt a_{c(\kappa-1)} \end{bmatrix}.$$

Notice that the condition $\bar{x}_e(0) = \bar{x}_e(\kappa)$ has been considered in the last row of $\bar{A}(c)$. The limit cycle trajectory is shown in Fig. 2, with $\bar{x}_e(0) = \bar{x}_e(\kappa) = x_e(0) = x_e(T)$.

Let the vector $\xi(t) = x(t) - x_e(t)$, the system (1) becomes

$$\begin{aligned} \dot{\xi}(t) &= A_{\sigma(t)}\xi(t) + B_{\sigma(t)}w(t) + l_{\sigma(t)}, \\ y_e(t) &= C_{\sigma(t)}\xi(t) + D_{\sigma(t)}w(t) \end{aligned} \quad (5)$$

where $l_{\sigma(t)} = A_{\sigma(t)}x_e(t) + a_{\sigma(t)} - \dot{x}_e(t)$ is the affine term, $y_e(t) = y(t) - C_{\sigma(t)}x_e(t)$.

Our goal is to determine the $\tau_{k(t)}$ corresponding to each $c[k(t)]$ to assure GES of the limit cycle $x_e(c)$. That is, we want to design a TFSS to guarantee that the origin $\xi(t) = 0$ of (5) be a GES equilibrium point, ensuring a appropriate upper limit of the H_∞ performance indexes.

The $\mathcal{I}^\kappa$ different switching sequences $c \in \mathcal{C}(\kappa)$ and the $\tau_{k(t)}$ of each $c[k(t)]$ allow us to define a family of candidate limit cycles

$$\mathcal{Y} = \{x_e(c) : c[k(t)] \in \mathcal{C}(\kappa)\}. \quad (6)$$

In order to screen out the desired limit cycle $x_e^* \in \mathcal{Y}$, the steady-state properties of the trajectory of system (1) are defined as screening conditions, such as oscillation frequency can be expressed as $\mathcal{Y}_e \subseteq \mathcal{Y}$. Given a reference point x^* , the selection criteria for the oscillation frequency is

$$\mathcal{Y}_e = \{x_e \in \mathcal{Y} : \frac{1}{\kappa} \sum_{i=0}^{\kappa-1} \|H(\bar{x}_e(i) - x^*)\|_2 < \rho\}, \quad (7)$$

which means the distance between the key node of $x_e(t)$ and x^* over $t \in [0, T]$ is smaller than ρ . The matrix $H \in \mathbb{R}^{1 \times n}$ is a weight assignment matrix provided by the designer, which reflects the designer's attention to the difference between the limit cycle state x_e and the reference point state x^* in each dimension. The positive parameter $\rho \in \mathbb{R}$ supplied by the designer is important in filtering the steady-state behavior of the system. When the ripple amplitude of the steady-state trajectory of system (1) is used as the screening condition, the following subsets can be used

$$\mathcal{Y}_e = \{x_e \in \mathcal{Y} : \max_{i \in \{0, 1, \dots, \kappa-1\}} \|H(\bar{x}_e(i) - x^*)\|_\infty < \varrho\}. \quad (8)$$

As with the former, ϱ is a parameter supplied by the designer. Note that the selection of the filter criteria is entirely dependent on the designer's interest in the steady-state behavior of the trajectory, and the subset $\mathcal{Y}_e$ can be defined in different forms.

Remark 2. To make readers better understand the research ideas of this paper, To apply the new TDSS conveniently, we propose the following Algorithm 1 to guide readers to ensure the GES of the desired limit cycle.

Algorithm 1 Operation Guide.

Step1: Determine κ and the desire switching sequence c .

Step2: According to the theorems presented in the next section, the dwell time $\tau_{k(t)}$ of the mode $c[k(t)]$ can be obtained, then a family of candidate limit cycles $\mathcal{Y}$ can be obtained by (6).

Step3: Take the desired limit cycle $\mathcal{X}_e^*$ from $\mathcal{Y}$.

Step4: Verify that the properties of the desired limit cycle $\mathcal{X}_e^*$ obtained in Step 3 whether meet the requirements such as (7) or (8). If the user's requirements for limit cycle performance are not met, repeat Step 3. If none of the desired limit cycle $\mathcal{X}_e^*$ in the family of candidate limit cycles $\mathcal{Y}$ meets the requirement, then skip to step 2 and readjust the dwell time $\tau_{k(t)}$.

Step5: The CTSAS (1) carries out periodic operation according to the switching sequence c obtained in Step 1 with the corresponding dwell time, the GES of the expected limit cycle $\mathcal{X}_e^*$ obtained in Step 3 can be guaranteed.

Before describing the main result, some definitions are given here.

Definition 1. For a given TDSS $\sigma(t)$ with the switching time t_{s_n} , $\Delta_{s_n} = t_{s_n} - t_{s_{n-1}}$ is the corresponding dwell time of s_n th switching. Then, the minimum dwell time $\tau_{\min}$ of $\sigma(t)$ is defined as $\tau_{\min} = \min_{s_n \in \mathbb{N}} \Delta_{s_n}$.

Definition 2. For a certain TDSS $\sigma(t)$, the equilibrium $\xi = 0$ of the switched affine system (5) (or the limit cycle $\mathcal{X}_e(c)$ of the switched affine system (1)) with $w(t) \equiv 0$ is GES if for any initial conditions $\xi(t_0)$, there exists $K > 0$, $\zeta < 0$ such that the solution of the system satisfies $\|\xi(t)\| \leq Ke^{\zeta(t-t_0)} \|\xi(t_0)\|$.

Definition 3. System (5) (or (1)) is GES and achieves the H_∞ performance with a weighted ℓ_2 -gain, if under zero initial condition, it holds that

(1) The equilibrium $\xi = 0$ of the switched affine system (5) (or the limit cycle $\mathcal{X}_e(c)$ of the switched affine system (1)) is GES when $w(t) \equiv 0$;

(2) For given scalars $\delta > 0$ and $\gamma > 0$, under zero initial condition, it holds that

$$\int_0^\infty e^{-\delta s} y_e^T(s) y_e(s) ds \leq \gamma^2 \int_0^\infty w^T(s) w(s) ds \quad (9)$$

for $\forall w(s) \neq 0 \in \ell_2[0, \infty)$.

3. Main results

In this section, by analyzing the stability of the CTSAS (5), the sufficient conditions with the TDSS will be designed to guarantee the GES of the desired limit cycle for the system (5). Besides, the conditions are generalized to deal with H_∞ control.

3.1. Stability analysis and control design

Before presenting the main theorems, the discretized Lyapunov function (DLF) constructed in this paper is introduced in advance. Construct the Lyapunov function as follows:

$$V_{\sigma(t)}(t) = \xi^T(t) P_{\sigma(t)}(t) \xi(t). \quad (10)$$

The DLF technique used in this work is given in detail as follows: We divide the interval $[t_{s_n}, t_{s_n} + \tau_{\min})$ into L segments described as $Q_{s_n, q} = [t_{s_n} + \phi_q, t_{s_n} + \phi_{q+1})$, $q = 0, 1, \dots, L-1$ of equal length $l = \frac{\tau_{\min}}{L}$, and then $\phi_q = ql = \frac{q\tau_{\min}}{L}$, $q = 0, 1, \dots, L$. The continuous matrix function $P_{\sigma(t)}(t)$, $t \in [t_{s_n}, t_{s_{n+1}})$ is chosen to be linear within each segment $Q_{s_n, q}$, $q = 0, 1, \dots, L-1$. Letting $P_{\sigma(t), q} = P_{\sigma(t)}(t_{s_n} + \phi_q)$, then since the matrix function $P_{\sigma(t)}(t)$ is piecewise linear in the minimal dwell time interval $[t_{s_n}, t_{s_n} + \tau_{\min})$, it can be expressed by a linear interpolation formula as the value at the dividing point, i.e., for $0 \leq \epsilon \leq 1$, $q = 0, 1, \dots, L-1$, and $t \in Q_{s_n, q}$, we have

$$P_{\sigma(t)}(t) = P_{\sigma(t)}(t + \phi_q + \epsilon l) = (1 - \epsilon)P_{\sigma(t), q} + \epsilon P_{\sigma(t), q+1} = P_{\sigma(t)}^q(\epsilon), \text{ where } \epsilon = \frac{t - t_{s_n} - \phi_q}{l}.$$

Thus, the discretized matrix function $P_{\sigma(t)}(t)$ can be described as

$$P_{\sigma(t)}(t) = \begin{cases} P_{\sigma(t)}^q(\epsilon), & t \in Q_{s_n, q}, q = 1, 2, \dots, L-1 \\ P_{\sigma(t), L}, & t \in [t_{s_n} + \tau_{\min}, t_{s_{n+1}}) \end{cases} \quad (11)$$

Theorem 1. Consider the CTSAS (5) with $w(t) \equiv 0$ and let the desired limit cycle sequence $c[k(t)]$ with κ terms be given. If there exist scalars $\alpha > 0$, $0 < \mu < 1$, $\tau_{\min} > 0$ and symmetric matrices $P_{c(p), q} > 0$, $p = 0, 1, \dots, \kappa-1$, $q = 0, 1, \dots, L$ such that

the following conditions

$$A_{c(p)}^T P_{c(p),q} + P_{c(p),q} A_{c(p)} + (P_{c(p),q+1} - P_{c(p),q}) \frac{L}{\tau_{\min}} - \alpha P_{c(p),q} < 0, \quad (12)$$

$$A_{c(p)}^T P_{c(p),q+1} + P_{c(p),q+1} A_{c(p)} + (P_{c(p),q+1} - P_{c(p),q}) \frac{L}{\tau_{\min}} - \alpha P_{c(p),q+1} < 0, \quad (13)$$

$$A_{c(p)}^T P_{c(p),L} + P_{c(p),L} A_{c(p)} - \alpha P_{c(p),L} < 0, \quad (14)$$

$$P_{c(p+1),0} - \mu P_{c(p),L} < 0, \quad (15)$$

hold for $\forall p \in \{0, 1, \dots, \kappa - 1\}$, $\forall q \in \{0, 1, \dots, L - 1\}$ and $c(\kappa) = c(0)$, then the dwell time τ_p of mode $c(p)$ and the period T of limit cycle sequence c satisfies

$$\begin{cases} \tau_{\min} \leq \tau_p, \\ T = \sum_{p=0}^{\kappa-1} \tau_p < -\frac{\kappa \ln \mu}{\alpha}, \end{cases} \quad (16)$$

assures the limit cycle $\mathcal{X}_e(c)$ is GES.

Proof. We choose the Lyapunov functional (10). By taking the derivative of $V_{\sigma(t)}(t)$ along the system trajectory, we have

$$\begin{aligned} \dot{V}_{\sigma(t)}(t) &= \dot{\xi}^T(t) P_{\sigma(t)}(t) \xi(t) + \xi^T(t) P_{\sigma(t)}(t) \dot{\xi}(t) + \xi^T(t) \dot{P}_{\sigma(t)}(t) \xi(t) \\ &= (A_{\sigma(t)} \xi(t) + l_{\sigma(t)})^T P_{\sigma(t)}(t) \xi(t) + \xi^T(t) P_{\sigma(t)}(t) (A_{\sigma(t)} \xi(t) + l_{\sigma(t)}) + \xi^T(t) \dot{P}_{\sigma(t)}(t) \xi(t) \\ &= (A_{c[k(t)]} \xi(t) + l_{c[k(t)]})^T P_{c[k(t)]}(t) \xi(t) + \xi^T(t) P_{c[k(t)]}(t) (A_{c[k(t)]} \xi(t) + l_{c[k(t)]}) + \xi^T(t) \dot{P}_{c[k(t)]}(t) \xi(t). \end{aligned} \quad (17)$$

Due to (2), we can get $l_{c[k(t)]} = 0$. Then

$$\dot{V}_{\sigma(t)}(t) = [A_{c[k(t)]} \xi(t)]^T P_{c[k(t)]}(t) \xi(t) + \xi^T(t) P_{c[k(t)]}(t) A_{c[k(t)]} \xi(t) + \xi^T(t) \dot{P}_{c[k(t)]}(t) \xi(t). \quad (18)$$

For $t \in Q_{s_n, q}$, the following equation can be derived:

$$\begin{aligned} \dot{P}_{c[k(t)]}(t) &= \dot{P}_{c[k(t)]}(t_{s_n} + \phi_q + \epsilon h) \\ &= \dot{P}_{c[k(t)]}^{(q)}(\epsilon) \\ &= (P_{c[k(t)],q+1} - P_{c[k(t)],q}) \dot{\epsilon}. \end{aligned} \quad (19)$$

Due to $\epsilon = \frac{t - t_{s_n} - \phi_q}{h}$, we have $\dot{\epsilon} = \frac{L}{\tau_{\min}}$. So,

$$\dot{P}_{c[k(t)]}(t) = (P_{c[k(t)],q+1} - P_{c[k(t)],q}) \frac{L}{\tau_{\min}}. \quad (20)$$

Without loss of generality, we assume $\forall t \in [t_{s_n}, t_{s_{n+1}})$, $s_n \in \mathbb{N}^+$.

Part 1: When $t \in [t_{s_n}, t_{s_n} + \tau_{\min})$, base on the Eqs. (18), (20), we have

$$\begin{aligned} \dot{V}_{\sigma(t)}(t) - \alpha V_{\sigma(t)}(t) &= [A_{c[k(t)]} \xi(t)]^T P_{c[k(t)]}(t) \xi(t) + \xi^T(t) P_{c[k(t)]}(t) A_{c[k(t)]} \xi(t) \\ &\quad + \xi^T(t) (P_{c[k(t)],q+1} - P_{c[k(t)],q}) \frac{L}{\tau_{\min}} \xi(t) - \alpha \xi^T(t) P_{c[k(t)]}(t) \xi(t) \\ &= \xi^T(t) \Xi_{c[k(t)],q} \xi(t), \end{aligned} \quad (21)$$

where

$$\Xi_{i,q} = A_{c[k(t)]}^T [(1 - \epsilon) P_{c[k(t)],q} + \epsilon P_{c[k(t)],q+1}] + [(1 - \epsilon) P_{c[k(t)],q} + \epsilon P_{c[k(t)],q+1}] A_i + (P_{c[k(t)],q+1} - P_{c[k(t)],q}) \frac{L}{\tau_{\min}} - \alpha [(1 - \epsilon) P_{c[k(t)],q} + \epsilon P_{c[k(t)],q+1}].$$

From (12) and (13), one can be obtained that $\dot{V}_{\sigma(t)}(t) - \alpha V_{\sigma(t)}(t) < 0$, $t \in [t_{s_n}, t_{s_n} + \tau_{\min})$.

Part 2: When $t \in [t_{s_n} + \tau_{\min}, t_{s_{n+1}})$, base on the Eq. (18), we see

$$\begin{aligned} \dot{V}_{\sigma(t)}(t) - \alpha V_{\sigma(t)}(t) &= [A_{c[k(t)]} \xi(t)]^T P_{c[k(t)],L} \xi(t) + \xi^T(t) P_{c[k(t)],L} A_{c[k(t)]} \xi(t) - \alpha \xi^T(t) P_{c[k(t)],L} \xi(t) \\ &= \xi^T(t) \Gamma_{c[k(t)],L} \xi(t), \end{aligned} \quad (22)$$

where $\Gamma_{c[k(t)],L} = A_{c[k(t)]}^T P_{c[k(t)],L} + P_{c[k(t)],L} A_{c[k(t)]} - \alpha P_{c[k(t)],L}$. From (14), one can be obtained that $\dot{V}_{\sigma(t)}(t) - \alpha V_{\sigma(t)}(t) < 0$, $t \in [t_{s_n} + \tau_{\min}, t_{s_{n+1}})$.

Thus, combining Part 1 and Part 2 lead to

$$\dot{V}_{\sigma(t)}(t) - \alpha V_{\sigma(t)}(t) < 0, \forall t \in [t_{s_n}, t_{s_{n+1}}). \quad (23)$$

Part 3: At the t_{s_n} switching moment, we have

$$\begin{aligned} & V_{\sigma(t_{s_n})}(t_{s_n}) - \mu V_{\sigma(t_{s_{n-1}})}(t_{s_n}^-) \\ &= \xi^T(t_{s_n})P_{c[k(t)]+1,0}\xi(t_{s_n}) - \mu \xi^T(t_{s_n}^-)P_{c[k(t)],L}\xi(t_{s_n}^-) \\ &= \xi^T(t_{s_n})(P_{c[k(t)]+1,0} - \mu P_{c[k(t)],L})\xi(t_{s_n}). \end{aligned} \quad (24)$$

Based on (15), we arrive at

$$V_{\sigma(t_{s_n})}(t_{s_n}) - \mu V_{\sigma(t_{s_{n-1}})}(t_{s_n}^-) < 0. \quad (25)$$

Then, from (23) and (25), we have

$$\begin{aligned} & V_{\sigma(t_{s_n})}(t) \\ & \leq e^{\alpha(t-t_{s_n})}V_{\sigma(t_{s_n})}(t_{s_n}) \\ & \leq \mu e^{\alpha(t-t_{s_n})}V_{\sigma(t_{s_{n-1}})}(t_{s_n}^-) \\ & \leq \mu e^{\alpha(t-t_{s_{n-1}})}V_{\sigma(t_{s_{n-1}})}(t_{s_{n-1}}^-) \\ & \leq \dots \\ & \leq \mu^{\kappa \frac{t_{s_n}-t_0}{T}} e^{\alpha(t-t_0)}V_{\sigma(t_0)}(t_0) \\ & = \mu^{-\kappa \frac{t-t_{s_n}}{T}} \mu^{\kappa \frac{t-t_0}{T}} e^{\alpha(t-t_0)}V_{\sigma(t_0)}(t_0) \\ & \leq \theta e^{\vartheta(t-t_0)}V_{\sigma(t_0)}(t_0), \end{aligned} \quad (26)$$

where $\theta = \mu^{-\kappa}$, $\vartheta = \frac{\kappa \ln \mu}{T} + \alpha$. It is clear that θ is a constant and (16) ensures that $\vartheta < 0$. Therefore, due to the Definition 3, under the dwell time $\tau_{k(t)}$ of mode $c[k(t)]$ and the period T of the limit cycle $\mathcal{X}_e(c)$ satisfies (16), the conditions of Theorem 1 are sufficient for GES of the origin $\xi = 0$ concerning system (5), or for the GES of the limit cycle $\mathcal{X}_e(c)$ regarding system (1). That's the end of the proof. $\square$

Remark 3. To the best of the authors' knowledge, this is the first work on GES of the desired limit cycle for CTSASs. There are some recent works [27,31,32] study the CTSAS with state-dependent switching function. The work [30] studies practical stability of CTSASs under minimum dwell time criteria while the work [33] under average dwell time criteria. In these references, the trajectories converge to a convergence region containing the desired equilibrium points. However, the system behavior within this region cannot be controlled, which results in the H_∞ performance index can't be taken into account. While in this work, ℓ_2 -gain and H_∞ control for a class of CTSASs will be investigated in Theorem 2. Some recent works [34,35] guarantee a local asymptotic stability of limit cycles by a Poincaré map method. Differently, the conditions of Theorem 1 are expressed in terms of LMIs, so it is easier to solve. Then, compared with the practical stability condition of the state-dependent switching function, the condition of Theorem 1 is less conservative than the time-dependent Lyapunov functional method as in [31], and a Lyapunov-Krasovskii functional method as in [32].

Remark 4. The very recently papers [36,37] have studied the global asymptotic stability of a limit cycle for CTSASs and DTSASs, respectively. However, the control strategies in these literatures are all state dependent, and H_∞ control is not considered in the reference [36]. In general, the TDSS designed in this paper is more convenient to be applied in practical projects because the system does not need to obtain the state information all the time.

3.2. H_∞ performance analysis and control design

Theorem 2. Consider the CTSAS (5) and let the desired limit cycle sequence $c[k(t)]$ with κ terms be given. If there exist scalars $\alpha > 0$, $0 < \mu < 1$, $\gamma > 0$, $\tau_{\min} > 0$ and symmetric matrices $P_{c(p),q} > 0$, $p = 0, 1, \dots, \kappa - 1$, $q = 0, 1, \dots, L$ such that the following conditions

$$\begin{bmatrix} \Psi_{c(p),q}^1(1,1) & \Psi_{c(p),q}^1(1,2) \\ * & D_{c(p)}^T D_{c(p)} - \gamma^2 \end{bmatrix} < 0, \quad (27)$$

$$\begin{bmatrix} \Psi_{c(p),q}^2(1,1) & \Psi_{c(p),q}^2(1,2) \\ * & D_{c(p)}^T D_{c(p)} - \gamma^2 \end{bmatrix} < 0, \quad (28)$$

$$\begin{bmatrix} \Psi_{c(p),L}^3(1,1) & \Psi_{c(p),q}^3(1,2) \\ * & D_{c(p)}^T D_{c(p)} - \gamma^2 \end{bmatrix} < 0, \quad (29)$$

$$P_{c(p+1),0} - \mu P_{c(p),L} < 0, \quad (30)$$

where

$$\Psi_{c(p),q}^1(1,1) = A_{c(p)}^T P_{c(p),q} + P_{c(p),q} A_{c(p)} + (P_{c(p),q+1} - P_{c(p),q}) \frac{L}{\tau_{\min}} - \alpha P_{c(p),q} + C_{c[k(t)]}^T C_{c(p)},$$

$\Psi_{c(p),q}^1(1,2) = P_{c(p),q} B_{c(p)} + C_{c(p)}^T D_{c(p)},$
 $\Psi_{c(p),q}^2(1,1) = A_{c(p)}^T P_{c(p),q+1} + P_{c(p),q+1} A_i + (P_{c(p),q+1} - P_{c(p),q}) \frac{L}{\tau_{\min}} - \alpha P_{c(p),q+1} + C_{c(p)}^T C_{c(p)},$
 $\Psi_{c(p),q}^2(1,2) = P_{c(p),q+1} B_{c(p)} + C_{c(p)}^T D_{c(p)},$
 $\Psi_{c(p),L}^3(1,1) = A_{c(p)}^T P_{c(p),L} + P_{c(p),L} A_{c(p)} - \alpha P_{c(p),L} + C_{c(p)}^T C_{c(p)},$
 $\Psi_{c(p),q}^3(1,2) = P_{c(p),L} B_{c(p)} + C_{c(p)}^T D_{c(p)},$
 hold for $\forall p \in \{0, 1, \dots, \kappa - 1\}$, $\forall q \in \{0, 1, \dots, L - 1\}$ and $c(\kappa) = c(0)$
 then the dwell time τ_p of mode $c(p)$ and the period T of limit cycle sequence c satisfies (16) assures the limit cycle $\mathcal{X}_e(c)$ is GES and the weighted H_∞ performance no greater than $\frac{\gamma}{\sqrt{e}}$.

Proof. Without loss of generality, we assume $\forall t \in [t_{s_n}, t_{s_{n+1}})$, $s_n \in \mathbb{N}^+$.

Part 1: When $t \in [t_{s_n}, t_{s_n} + \tau_{\min})$, by the Eqs. (18), (20), we have

$$\begin{aligned}
 & \dot{V}_{\sigma(t)}(t) - \alpha V_{\sigma(t)}(t) + R(t) \\
 &= [A_{c[k(t)]} \xi(t) + B_{c[k(t)]} w(t)]^T P_{c[k(t)],q}(t) \xi(t) \\
 & \quad + \xi^T(t) P_{c[k(t)],q}(t) [A_{c[k(t)]} \xi(t) + B_{c[k(t)]} w(t)] \\
 & \quad + \xi^T(t) (P_{c[k(t)],q+1} - P_{c[k(t)],q}) \frac{L}{\tau_{\min}} \xi(t) \\
 & \quad - \alpha \xi^T(t) P_{c[k(t)],q}(t) \xi(t) \\
 & \quad + [C_{c[k(t)]} \xi(t) + D_{c[k(t)]} w(t)]^T \\
 & \quad \times [C_{c[k(t)]} \xi(t) + D_{c[k(t)]} w(t)] \\
 & \quad - \gamma^2 w^T(t) w(t) \\
 &= \begin{bmatrix} \xi(t) \\ w(t) \end{bmatrix}^T \begin{bmatrix} \Pi_{c[k(t)],q}^{(1,1)} & \Pi_{c[k(t)],q}^{(1,2)} \\ * & \Pi_{c[k(t)],q}^{(2,2)} \end{bmatrix} \begin{bmatrix} \xi(t) \\ w(t) \end{bmatrix},
 \end{aligned} \tag{31}$$

where

$$\begin{aligned}
 & R(t) = y_e^T(t) y_e(t) - \gamma^2 w^T(t) w(t), \\
 & \Pi_{c[k(t)],q}^{(1,1)} = A_{c[k(t)]}^T [(1 - \epsilon) P_{c[k(t)],q} + \epsilon P_{c[k(t)],q+1}] + (P_{c[k(t)],q+1} - P_{c[k(t)],q}) \frac{L}{\tau_{\min}} - \alpha [(1 - \epsilon) P_{c[k(t)],q} + \epsilon P_{c[k(t)],q+1}] + \\
 & C_{c[k(t)]}^T C_{c[k(t)]}, \\
 & \Pi_{c[k(t)],q}^{(1,2)} = [(1 - \epsilon) P_{c[k(t)],q} + \epsilon P_{c[k(t)],q+1}] B_{c[k(t)]} + C_{c[k(t)]}^T D_{c[k(t)]}, \\
 & \Pi_{c[k(t)],q}^{(2,2)} = D_{c[k(t)]}^T D_{c[k(t)]} - \gamma^2 I.
 \end{aligned}$$

From (27), (28), one can be obtained that $\dot{V}_{\sigma(t)}(t) - \alpha V_{\sigma(t)}(t) + R(t) < 0$, $t \in [t_{s_n}, t_{s_n} + \tau_{\min})$.

Part 2: When $t \in [t_{s_n} + \tau_{\min}, t_{s_{n+1}})$, we see

$$\begin{aligned}
 & \dot{V}_{\sigma(t)}(t) - \alpha V_{\sigma(t)}(t) + R(t) \\
 &= [A_{c[k(t)]} \xi(t) + B_{c[k(t)]} w(t)]^T P_{c[k(t)],L} \xi(t) \\
 & \quad + \xi^T(t) P_{c[k(t)],L} [A_{c[k(t)]} \xi(t) + B_{c[k(t)]} w(t)] \\
 & \quad + [C_{c[k(t)]} \xi(t) + D_{c[k(t)]} w(t)]^T \\
 & \quad \times [C_{c[k(t)]} \xi(t) + D_{c[k(t)]} w(t)] \\
 & \quad - \alpha \xi^T(t) P_{c[k(t)],L} \xi(t) - \gamma^2 w^T(t) w(t) - \beta \\
 &= \begin{bmatrix} \xi(t) \\ w(t) \\ 1 \end{bmatrix}^T \begin{bmatrix} \Upsilon_{c[k(t)],L}^{(1,1)} & \Upsilon_{c[k(t)],L}^{(1,2)} \\ * & \Upsilon_{c[k(t)],L}^{(2,2)} \end{bmatrix} \begin{bmatrix} \xi(t) \\ w(t) \\ 1 \end{bmatrix},
 \end{aligned} \tag{32}$$

where

$$\begin{aligned}
 & \Upsilon_{c[k(t)],L}^{(1,1)} = A_{c[k(t)]}^T P_{c[k(t)],L} + P_{c[k(t)],L} A_{c[k(t)]} - \alpha P_{c[k(t)],L} + C_{c[k(t)]}^T C_{c[k(t)]}, \\
 & \Upsilon_{c[k(t)],L}^{(1,2)} = P_{c[k(t)],L} B_{c[k(t)]} + C_{c[k(t)]}^T D_{c[k(t)]}, \\
 & \Upsilon_{c[k(t)],L}^{(2,2)} = D_{c[k(t)]}^T D_{c[k(t)]} - \gamma^2 I.
 \end{aligned}$$

From (29), one can be obtained that $\dot{V}_{\sigma(t)}(t) - \alpha V_{\sigma(t)}(t) + R(t) < 0$, $t \in [t_{s_n} + \tau_{\min}, t_{s_{n+1}})$.

Thus, combining Part 1 and Part 2 lead to

$$\dot{V}_{\sigma(t)}(t) - \alpha V_{\sigma(t)}(t) + R(t) < 0, \forall t \in [t_{s_n}, t_{s_{n+1}}). \tag{33}$$

Part 3: At the t_{s_n} switching moment, (24) and (30) ensure

$$V_{\sigma(t_{s_n})}(t_{s_n}) - \mu V_{\sigma(t_{s_{n-1}})}(t_{s_n}^-) < 0. \tag{34}$$

Base on (33) and (34), we can obtain

$$\begin{aligned}
 & V_{\sigma(s_n)}(t) \\
 & \leq e^{\alpha(t-t_{s_n})} V_{\sigma(s_n)}(t_{s_n}) - \int_{t_{s_n}}^t e^{\alpha(t-\varsigma)} R(\varsigma) d\varsigma \\
 & \leq \mu e^{\alpha(t-t_{s_n})} V_{\sigma(s_{n-1})}(t_{s_n}^-) - \int_{t_{s_n}}^t e^{\alpha(t-\varsigma)} R(\varsigma) d\varsigma \\
 & \leq \mu e^{\alpha(t-t_{s_{n-1}})} V_{\sigma(s_{n-1})}(t_{s_{n-1}}^-) - \int_{t_{s_{n-1}}}^t \mu e^{\alpha(t-\varsigma)} R(\varsigma) d\varsigma - \int_{t_{s_n}}^t e^{\alpha(t-\varsigma)} R(\varsigma) d\varsigma \\
 & \leq \dots \\
 & \leq \mu^\kappa e^{\alpha(t-t_0)} V_{\sigma(t_0)}(t_0) - \sum_{l=1}^n \int_{t_{s_{l-1}}}^{t_{s_l}} \mu^{N(\varsigma,t)} e^{\alpha(t-\varsigma)} R(\varsigma) d\varsigma - \int_{t_{s_n}}^t e^{\alpha(t-\varsigma)} R(\varsigma) d\varsigma \\
 & \leq \mu^{-\kappa} e^{\alpha(t-t_0)} V_{\sigma(t_0)}(t_0) - \sum_{l=1}^n \int_{t_{s_{l-1}}}^{t_{s_l}} e^{N(\varsigma,t) \ln \mu + \alpha(t-\varsigma)} R(\varsigma) d\varsigma - \int_{t_{s_n}}^t e^{\alpha(t-\varsigma)} R(\varsigma) d\varsigma \\
 & \leq \theta e^{\vartheta(t-t_0)} V_{\sigma(t_0)}(t_0) - \Theta(s),
 \end{aligned} \tag{35}$$

where $N(\varsigma, t)$ denote the number of switching over (ς, t) . $\theta = \mu^{-\kappa}$, $\vartheta = \frac{\kappa \ln \mu}{T} + \alpha$, $\Theta(s) = \sum_{l=1}^n \int_{t_{s_{l-1}}}^{t_{s_l}} e^{N(\varsigma,t) \ln \mu + \alpha(t-\varsigma)} R(\varsigma) d\varsigma + \int_{t_{s_n}}^t e^{\alpha(t-\varsigma)} R(\varsigma) d\varsigma$.

Under zero initial condition, it follows from (35) that

$$\begin{aligned}
 & \sum_{l=1}^n \int_{t_{s_{l-1}}}^{t_{s_l}} e^{N(\varsigma,t) \ln \mu + \alpha(t-\varsigma)} y_e^T(\varsigma) y_e(\varsigma) d\varsigma + \int_{t_{s_n}}^t e^{\alpha(t-\varsigma)} y_e^T(\varsigma) y_e(\varsigma) d\varsigma \\
 & \leq \sum_{l=1}^n \int_{t_{s_{l-1}}}^{t_{s_l}} e^{N(\varsigma,t) \ln \mu + \alpha(t-\varsigma)} \gamma^2 w^T(\varsigma) w(\varsigma) d\varsigma + \int_{t_{s_n}}^t e^{\alpha(t-\varsigma)} \gamma^2 w^T(\varsigma) w(\varsigma) d\varsigma.
 \end{aligned} \tag{36}$$

Then, multiplying both sides of (36) by $e^{-N(0,t) \ln \mu}$, we have

$$\tilde{\Theta}(s) \leq \hat{\Theta}(s). \tag{37}$$

where

$$\begin{aligned}
 \tilde{\Theta}(s) &= \sum_{l=1}^n \int_{t_{s_{l-1}}}^{t_{s_l}} e^{-N(0,\varsigma) \ln \mu + \alpha(t-\varsigma)} y_e^T(\varsigma) y_e(\varsigma) d\varsigma \\
 &+ \int_{t_{s_n}}^t e^{-N(0,t) \ln \mu + \alpha(t-\varsigma)} y_e^T(\varsigma) y_e(\varsigma) d\varsigma,
 \end{aligned} \tag{38}$$

$$\begin{aligned}
 \hat{\Theta}(s) &= \sum_{l=1}^n \int_{t_{s_{l-1}}}^{t_{s_l}} e^{-N(0,\varsigma) \ln \mu + \alpha(t-\varsigma)} \gamma^2 w^T(\varsigma) w(\varsigma) d\varsigma \\
 &+ \int_{t_{s_n}}^t e^{-N(0,t) \ln \mu + \alpha(t-\varsigma)} \gamma^2 w^T(\varsigma) w(\varsigma) d\varsigma.
 \end{aligned} \tag{39}$$

Owing to $0 < \mu < 1$, we can have

$$\begin{aligned}
 \tilde{\Theta}(s) &\geq \sum_{l=1}^n \int_{t_{s_{l-1}}}^{t_{s_l}} e^{\alpha(t-\varsigma)} y_e^T(\varsigma) y_e(\varsigma) d\varsigma + \int_{t_{s_n}}^t e^{\alpha(t-\varsigma)} y_e^T(\varsigma) y_e(\varsigma) d\varsigma \\
 &= \int_0^t e^{\alpha(t-\varsigma)} y_e^T(\varsigma) y_e(\varsigma) d\varsigma.
 \end{aligned} \tag{40}$$

Notice that $N(0, t) \leq \frac{t+T}{T}$, together with (16), we have

$$\begin{aligned}
 \hat{\Theta}(s) &\leq \sum_{l=1}^n \int_{t_{s_{l-1}}}^{t_{s_l}} e^{-\kappa \frac{\varsigma+T}{T} \ln \mu + \alpha t} \gamma^2 w^T(\varsigma) w(\varsigma) d\varsigma + \int_{t_{s_n}}^t e^{-\kappa \frac{\varsigma+T}{T} \ln \mu + \alpha(t)} \gamma^2 w^T(\varsigma) w(\varsigma) d\varsigma \\
 &= \int_0^t e^{-\kappa \frac{\varsigma+T}{T} \ln \mu + \alpha t} \gamma^2 w^T(\varsigma) w(\varsigma) d\varsigma.
 \end{aligned} \tag{41}$$

Combining (40) and (41) leads to

$$\begin{aligned}
 & \int_0^t e^{\alpha(t-\varsigma)} y_e^T(\varsigma) y_e(\varsigma) d\varsigma \leq \int_0^t e^{-\kappa \frac{\varsigma+T}{T} \ln \mu + \alpha t} \gamma^2 w^T(\varsigma) w(\varsigma) d\varsigma \\
 & \iff \\
 & \int_0^t e^{-(\alpha - \frac{\kappa \ln \mu}{T}) \varsigma} y_e^T(\varsigma) y_e(\varsigma) d\varsigma \leq \int_0^t \frac{\gamma^2}{e} w^T(\varsigma) w(\varsigma) d\varsigma
 \end{aligned} \tag{42}$$

Let $t_0 = 0$ and integrate both sides of this inequality form $t_0 = 0$ to ∞ lead to

$$\int_0^\infty e^{-\delta \varsigma} y_e^T(\varsigma) y_e(\varsigma) d\varsigma \leq \int_0^\infty \frac{\gamma^2}{e} w^T(\varsigma) w(\varsigma) d\varsigma. \tag{43}$$

where $\delta = \alpha - \frac{\kappa \ln \mu}{T} > 0$. That's the end of the proof. $\square$

Theorem 2 provides sufficient conditions for the H_∞ control design of CTSASs. These conditions are LMIs over all variables. Therefore, for given parameters (α and μ), when γ is given, **Theorem 2** provides the method for the H_∞ control design of CTSASs. When γ is unknown, the minimum (optimal) weighting H_∞ performance indicator γ can be obtained by the following theorem and solved as an LMI optimization problem.

Theorem 3. Consider the CTSAS (5) and let the desired limit cycle sequence $c[k(t)]$ with κ terms be given. If there exist scalars $\alpha > 0$, $0 < \mu < 1$, $\gamma > 0$, $\tau_{\min} > 0$ and symmetric matrices $P_{c(p),q} > 0$, $p = 0, 1, \dots, \kappa - 1$, $q = 0, 1, \dots, L$ satisfying

$$\begin{aligned}
 & \min_{\alpha, \mu, P_{c(p),q}} \gamma, \\
 & \text{s.t. (27) - (30),}
 \end{aligned} \tag{44}$$

then the dwell time τ_p of mode $c(p)$ and the period T of limit cycle sequence c satisfies (16) assures the limit cycle $\mathcal{X}_c(c)$ is GES and the weighted H_∞ performance no greater than $\frac{\gamma}{\sqrt{e}}$.

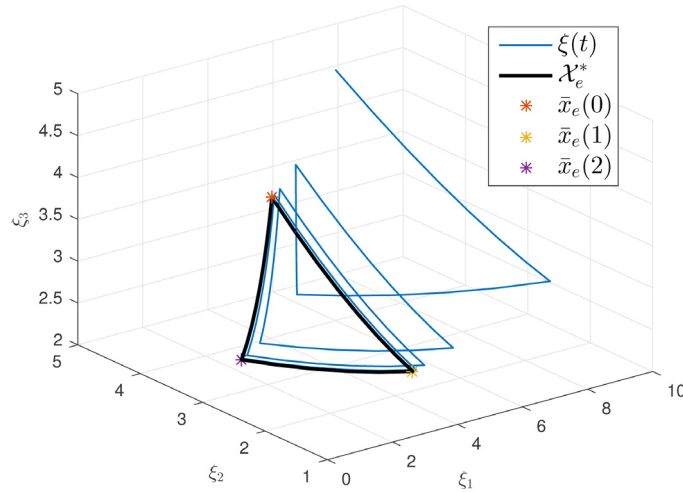


Fig. 3. The limit cycle and trajectories of $\xi(t)$.

4. Simulation

In this section, a numerical example and a practical example are provided to verify those main results got in this technical note.

Example 1 (Numerical example): Consider a CTSAS with three unstable subsystems, and the parameters of each subsystem are as follows:

$$A_1 = \begin{bmatrix} 0.1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, A_2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0.1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, A_3 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0.1 \end{bmatrix}, a_1 = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, a_2 = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}, a_3 = \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix}.$$

The control goal of this example is to keep the values of the first two state components around $x^* = [3 \ 4 \ \bullet]^T$ while the system is in a steady-state, where $\bullet$ represents an unimportant value. Select $\kappa = 2$ associated with desire switching sequence $c = (1, 2)$. Set $\alpha = 0.4$, $\mu = 0.6$, $L = 1$, $\tau_{\min} = 0.5$, $\kappa = 3$, one can be obtained that $-\kappa \frac{\ln \mu}{\alpha} = 2.1893$ base on (16). The dwell time of the switching sequence c can be chosen as $\tau_0 = 0.7$, $\tau_1 = 0.5$, $\tau_2 = 0.8$, then we can obtain the following feasible solution with the Theorem 1.

$$P_{10} = \begin{bmatrix} 11.4085 & 0 & 0 \\ 0 & 8.3729 & 0 \\ 0 & 0 & 8.3729 \end{bmatrix}, P_{11} = \begin{bmatrix} 14.2067 & 0 & 0 \\ 0 & 19.3521 & 0 \\ 0 & 0 & 19.3521 \end{bmatrix}, P_{20} = \begin{bmatrix} 8.3729 & 0 & 0 \\ 0 & 11.4085 & 0 \\ 0 & 0 & 8.3729 \end{bmatrix}, \\ P_{21} = \begin{bmatrix} 19.3521 & 0 & 0 \\ 0 & 14.2067 & 0 \\ 0 & 0 & 19.3521 \end{bmatrix}, P_{30} = \begin{bmatrix} 8.3729 & 0 & 0 \\ 0 & 8.3729 & 0 \\ 0 & 0 & 11.4085 \end{bmatrix}, P_{31} = \begin{bmatrix} 19.3521 & 0 & 0 \\ 0 & 19.3521 & 0 \\ 0 & 0 & 14.2067 \end{bmatrix}.$$

Based on the dwell time τ_0, τ_1, τ_2 with the switching sequence c , the key nodes $\bar{x}_e(0) = [0.4579 \ 2.1279 \ 4.7002]^T$, $\bar{x}_e(1) = [2.6550 \ 1.0178 \ 2.7698]^T$ and $\bar{x}_e(2) = [2.3868 \ 3.6202 \ 2.0450]^T$ in $x_e(t)$ can be obtained from (4). So, the desired limit cycle $\mathcal{X}_e^*$ can be obtain by (6). Setting $H = \text{diag}(1, 1, 0)$, $\rho = \varrho = 3$, then we can obtain $\frac{1}{\kappa} \sum_{i=0}^{\kappa-1} \|H(\bar{x}_e(i) - x^*)\|_2 = 2.2935$ and $\max_{i \in \{0, \dots, \kappa-1\}} \|H(\bar{x}_e(i) - x^*)\|_\infty = 2.9822$ from (7) and (8), respectively. Driven by the switching sequence c and dwell time τ_0, τ_1, τ_2 , the ripple amplitude and oscillation frequency of the desired limit cycle $\mathcal{X}_e^*$ meet the requirements.

Set the trajectory $x(t)$ starts from $[6 \ 4 \ 5]^T$ then the state trajectories $x(t)$, the limit cycle $\mathcal{X}_e^*$ and three key nodes are shown in Fig. 3. As can be seen from the Fig. 3, the sequence $x(t)$ is attracted by the limit cycle $\mathcal{X}_e^*$. Once state $x(t)$ attain the limit cycle $\mathcal{X}_e^*$, it is confined within the attractive sets. Figure 4 shows the time evolution of the state response $\xi(t)$ with the corresponding TDSS $\sigma(t)$, where the state of the system approached zero driven by the switching signal. The simulation results are shown in Figs. 3 and 4 demonstrate the effectiveness of the proposed switching strategy.

Example 2 (Practical example):

This example was borrowed from Sanchez et al. [27], which treats the practical stability of operating points for CTSASs. The boost converter is depicted as Fig. 5 and the system parameters are $E = 100V$, $C = 470\mu F$, $L = 500\mu H$, $R_1 = 2\Omega$ and $R_2 = 50\Omega$. The state variable is $x(t) = [i_L(t) \ v_C(t)]^T$. The system matrices can be described as

$$A_1 = \begin{bmatrix} -\frac{R_1}{L} & 0 \\ 0 & -\frac{1}{R_2 C} \end{bmatrix}, A_2 = \begin{bmatrix} -\frac{R_1}{L} & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{R_2 C} \end{bmatrix}, a_1 = a_2 = \begin{bmatrix} \frac{E}{L} \\ 0 \end{bmatrix}, B_1 = B_2 = \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix}, C_1 = C_2 = \text{diag}(1, 1),$$

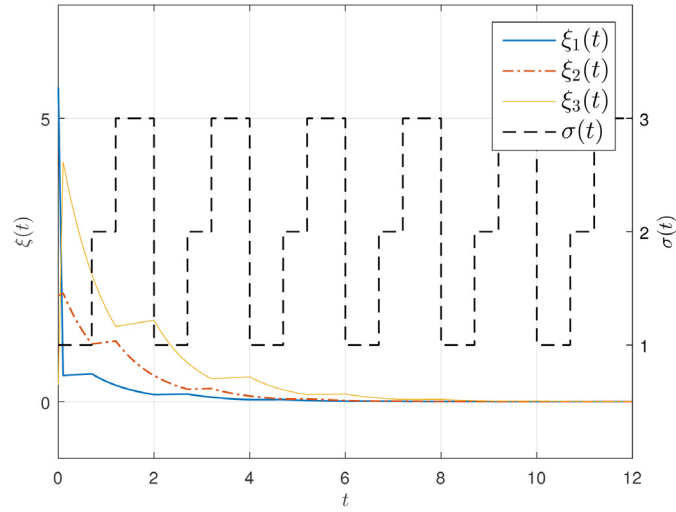


Fig. 4. The state response with the time-dependent switching signal.

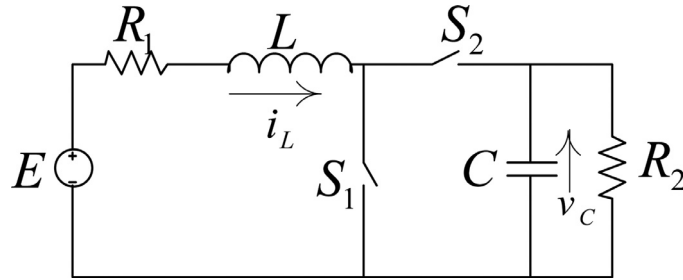


Fig. 5. Schematic of boost converter.

and $D_1 = D_2 = [0 \ 0 \ 0]^T$.

Now our goal is to design a TDSS $\sigma(t)$ with the switching sequence c and dwell time to mitigate the effects of external inputs such as voltage oscillations. In contrast with the results of [27], which guarantee only practical stability, the method proposed in this paper not only ensures GES of the desire limit cycle X_e^* but also guarantees H_∞ performance indexes.

The control goal of this example is to keep the values of the scend state components around $x^* = [\bullet \ 180]^T$ while the system is in a steady-state. Select $\kappa = 2$ associated with desire switching sequence $c = (1, 2)$. Set $L = 2$, $\tau_{\min} = 5 \times 10^{-4}$, $\kappa = 2$, one can be obtained that $\gamma = 19.0$ base on Theorem 3. The dwell time of the switching sequence c can be choosed as $\tau_0 = \tau_1 = \tau_{\min}$. Based on the dwell time τ_0, τ_1 with the switching sequence c , the key nodes $\bar{x}_e(0) = [-31.4891 \ 167.6420]^T$ and $\bar{x}_e(1) = [43.6634 \ 167.6420]^T$ in $x_e(t)$ can be obtained from (4). So, the desired limit cycle $\mathcal{X}_e^*$ can be obtain by (6). Setting $H = \text{diag}(0, 0.5)$, $\rho = \varrho = 7$, then we can obtain $\frac{1}{\kappa} \sum_{i=0}^{\kappa-1} \|H(\bar{x}_e(i) - x^*)\|_2 = 6.1790$ and $\max_{i \in \{0, \dots, \kappa-1\}} \|H(\bar{x}_e(i) - x^*)\|_\infty = 6.1790$ from (7) and (8), respectively. Driven by the switching sequence c and dwell time τ_0, τ_1 , the ripple amplitude and oscillation frequency of the desired limit cycle $\mathcal{X}_e^*$ meet the requirements.

Considering the following exogenous input

$$w(t) = \begin{cases} 120 \sin(\pi t), & t \in [0.015, 0.02) \\ 0, & \text{otherwise} \end{cases} \quad (45)$$

The initial conditions $x_0 = [-1.5 \ 4]^T$, then the system state vector $\xi(t)$, the limit cycle $\mathcal{X}_e^*$ and two key nodes are shown in Fig. 6. It is clear from the Fig. 6 that the sequence $x(t)$ is attracted by the limit cycle $\mathcal{X}_e^*$. Once state $x(t)$ attain the limit cycle $\mathcal{X}_e^*$, it is confined within the attractive sets. Figure 7 shows the time evolution of the state response $\xi(t)$ with the corresponding TDSS $\sigma(t)$, where the state of the system approached zero driven by the switching signal. Figure 7 demonstrates that the TDSS designed in this paper can successfully suppress the effects of external disturbance $w(t)$ on the system state. The simulation results are shown in Figs. 6 and 7 demonstrate the effectiveness of the proposed switching strategy.

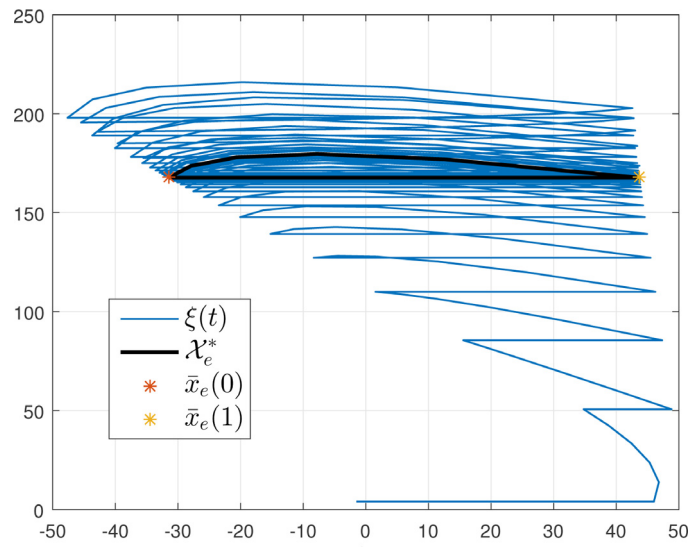


Fig. 6. The limit cycle and trajectories of $\xi(t)$.

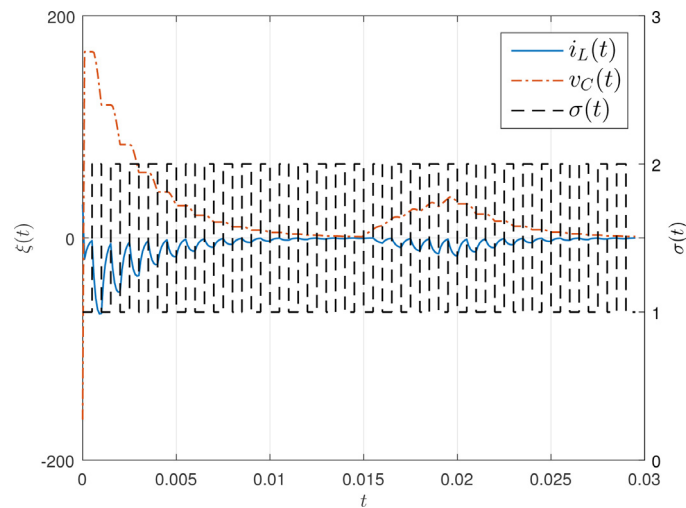


Fig. 7. The state response with the time-dependent switching signal.

5. Conclusion

In this technical note, the problem of H_∞ control to assuring the GES of the desired limit cycle under time-dependent switching signal for CTSASs has been solved. Sufficient conditions have been obtained by using the stabilization property of switching behavior and the DLF technique. Our method provides a TDSS to exploit the stabilization role of switching behaviors, thus it has further application value than the previous results via state-dependent switching signal. Finally, a numerical example and a practical example are given to illustrate the validity of the proposed theorems. The method presented in this paper can be easily extended to the sampling control of a CTSAS, which will provide a direction for designing the H_∞ sampling controller for a CTSAS.

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